

Sunil Tiwari

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EDUCATION

University of Utah

Master of Science in Cybersecurity Management (Specialization: Agentic AI in SecOps)

Salt Lake City, UT

Expected July 2026

Utah Valley University

Bachelor of Science in Information Technology (Network Security Focus)

Orem, UT

December 2022

TECHNICAL SKILLS

Agentic AI & Orchestration: CrewAI | LangChain | LLM SecOps | Prompt Engineering | Llama 3.2 | Ollama | RAG

Security Engineering: Autonomous Incident Response | SIEM (Splunk/Wazuh) | Forensic Triage | MITRE ATT&CK | Linux Kernel Networking | Vulnerability Management

Infrastructure & DevOps: NVIDIA DGX (ARM64) | Python (FastAPI) | Docker | Azure | Active Directory | PowerShell | Terraform

GRADUATE RESEARCH & SECURITY INITIATIVES

Hybrid-LLM Autonomous SOC & Active Defense Framework

December 2025 – Present

Lead Architect (MS Capstone Research)

- Architected a Hybrid-LLM Autonomous SOC utilizing the CrewAI framework to orchestrate local edge models (Llama 3.2) and cloud intelligence (GPT-4o) for zero-intervention threat triage and incident response.
- Integrated isolated Docker honeypots with a custom FastAPI backend, engineering regex-driven Python telemetry bridges to dynamically extract ephemeral attacker IPs and ingest raw log data into the cognitive pipeline.
- Engineered a deterministic Active Defense Governance Layer in Python to intercept and audit AI-generated actuation commands, enforcing strict infrastructure IP whitelisting to prevent AI hallucinations and host-level Self-Denial-of-Service (DoS).
- Ported the agentic engine to defend a simulated multi-tenant Web Application stack, reconfiguring RAG playbooks and decoders to handle Layer 7 telemetry and triage web-specific vulnerabilities (SQLi/RCE).
- Achieved autonomous threat containment by programmatically actuating dynamic iptables drop rules against validated malicious endpoints, preserving core DNS and network routing integrity.

Secure LLMOps Gateway (Stateless Security Proxy)

May 2026

SecOps & Platform Developer

- Engineered a stateless, high-performance security proxy in Python (FastAPI) designed to intercept, sanitize, and audit Generative AI prompt payloads prior to external routing.
- Designed a high-precision Data Loss Prevention (DLP) engine using regular expressions to auto-redact PII (Social Security Numbers and Credit Card details) on-the-fly, enforcing strict compliance with PCI-DSS and SOC 2 data-in-motion standards.
- Overrode LLM safety-refusal guardrails on redacted payloads by developing a custom context preservation layer in system instructions, allowing local Llama 3.2 models to process sanitized text without false-positive blocks.
- Implemented SIEM-ready JSON logging to stdout that tracks processing latency, transaction UUIDs, and detection flags while completely preventing raw-text PII leakage to downstream log collectors.
- Hardened application delivery by packaging the proxy inside a lightweight Docker container configured to run under a restricted non-root system user (appuser UID 10001) to eliminate container breakout risks.

PROFESSIONAL EXPERIENCE

University of Utah

Graduate Security Researcher

Salt Lake City, UT

May 2025 – Present

- Designing secure local LLM inference pipelines using FastAPI and Docker to ensure privacy-compliant AI deployments on enterprise hardware.
- Simulating network intrusion scenarios to research and validate automated AI-driven active defense mitigation playbooks.

Karuwaa Express

IT Operations & Security Manager

Sandy, UT

August 2024 – April 2025

- Executed a Formal Risk Treatment Plan for high-risk assets and implemented Role-Based Access Control (RBAC).
- Designed mandatory Staff Security Awareness Training, achieving a 0% phishing click rate.
- Mitigated internal fraud through least privilege enforcement and comprehensive security monitoring.

Intermountain Health

Patient Access Specialist (Seasonal Contract)

UT

May 2024 – August 2024

- Validated patient identity and secured highly sensitive Protected Health Information (PHI) under HIPAA and PCI-DSS compliance frameworks.
- Identified and escalated potential identity fraud anomalies during high-volume digital intake processes.

Utah Valley University

Orem, UT

Network Security Lab Assistant

September 2021 – December 2022

- Configured sandboxed laboratory environments using virtualization tools to simulate network security exploits and defensive exercises.
- Assisted in maintaining local Active Directory endpoints and resolving access-related system misconfigurations.

Walmart eCommerce

Remote

Partner Support Specialist

September 2020 – September 2021

- Managed enterprise workflows across Salesforce instances, resolving high-priority technical and partner issues.
- Drove process optimization resulting in a 20% improvement in overall team productivity.

CenturyLink

Monroe, LA

IT Support Help Desk

January 2019 – May 2019

- Managed Active Directory for 100+ endpoints, enforcing GPOs and resolving Access Control List (ACL) violations.
- Provided remote systems support and integration troubleshooting for VPN and Citrix infrastructure using PowerShell.

CERTIFICATIONS

CompTIA CySA+ | CompTIA SecAI+ (MSCM Capstone Candidate)